

Geospatial Data Analysis — Spatial Expansion Report

Important data note: No source regional dataset was provided for this task. To demonstrate the complete workflow, this submission uses a clearly labeled illustrative India city-level dataset with State, City, Postal Code, users, orders, revenue, and coordinates. The findings are therefore methodology/demo results, not real business findings.

Executive Summary

The illustrative dataset contains 15 city-region records, 13,020 users, 2,790 orders, and █4.58 crore in revenue. An underserved-region proxy was calculated from a demand index (users + 2×orders) relative to current revenue. The three highest-scoring regions are Patna, Bihar, Nagpur, Maharashtra, Lucknow, Uttar Pradesh.

Core Metrics

Regions	15
Users	13,020
Orders	2,790
Revenue	█45,800,000
Average revenue/user	█3,419

Regional Aggregation

State	Cities	Users	Orders	Revenue	Revenue/User	Underserved Score
Maharashtra	3	2,550	530	█8.45M	█3,314	427.22
Karnataka	2	2,000	440	█7.85M	█3,925	366.88
Gujarat	2	2,000	430	█7.10M	█3,550	402.82
Tamil Nadu	1	1,250	280	█4.80M	█3,840	377.08
Telangana	1	1,050	230	█3.90M	█3,714	387.18
Delhi	1	1,000	220	█3.60M	█3,600	400.00
West Bengal	1	850	170	█2.80M	█3,294	425.00
Uttar Pradesh	1	750	155	█2.35M	█3,133	451.06
Madhya Pradesh	1	600	125	█1.95M	█3,250	435.90
Rajasthan	1	550	115	█1.75M	█3,182	445.71
Bihar	1	420	95	█1.25M	█2,976	488.00

Top 3 High-Potential Underserved Regions

The ranking below is generated only from the illustrative proxy score. It should be replaced with the same calculation on the actual business dataset.

Rank	City	State	Users	Orders	Revenue	Revenue/User	Score
1	Patna	Bihar	420	95	■1,250,000	■2,976	488.00
2	Nagpur	Maharashtra	650	120	■1,850,000	■2,846	481.08
3	Lucknow	Uttar Pradesh	750	155	■2,350,000	■3,133	451.06

Geographic Visualization

The accompanying interactive Folium HTML map plots each city using latitude/longitude, scales marker size by revenue, and highlights the top three opportunity regions.

Expansion Interpretation

- Use the top-scoring regions as investigation candidates rather than automatic expansion decisions.
- Validate local demand with market size, competition, logistics/serviceability, customer acquisition cost, and margin before investment.
- If the real dataset contains repeated customer transactions, aggregate at State/City/Postcode level and recalculate the same demand and underserved metrics.
- For a true choropleth, use state/district boundary GeoJSON and join the aggregated metrics by a stable geographic identifier.

Limitations

The supplied task did not include a dataset, so all values, coordinates, and opportunity findings in this demo are illustrative. The underserved score is a transparent proxy, not a validated market-potential model. Postal codes are identifiers here and are not used as precise geocoding evidence.